



Automata Formal Languages & Logic

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Department of Computer Science & Engineering

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Applications of Finite State Machines

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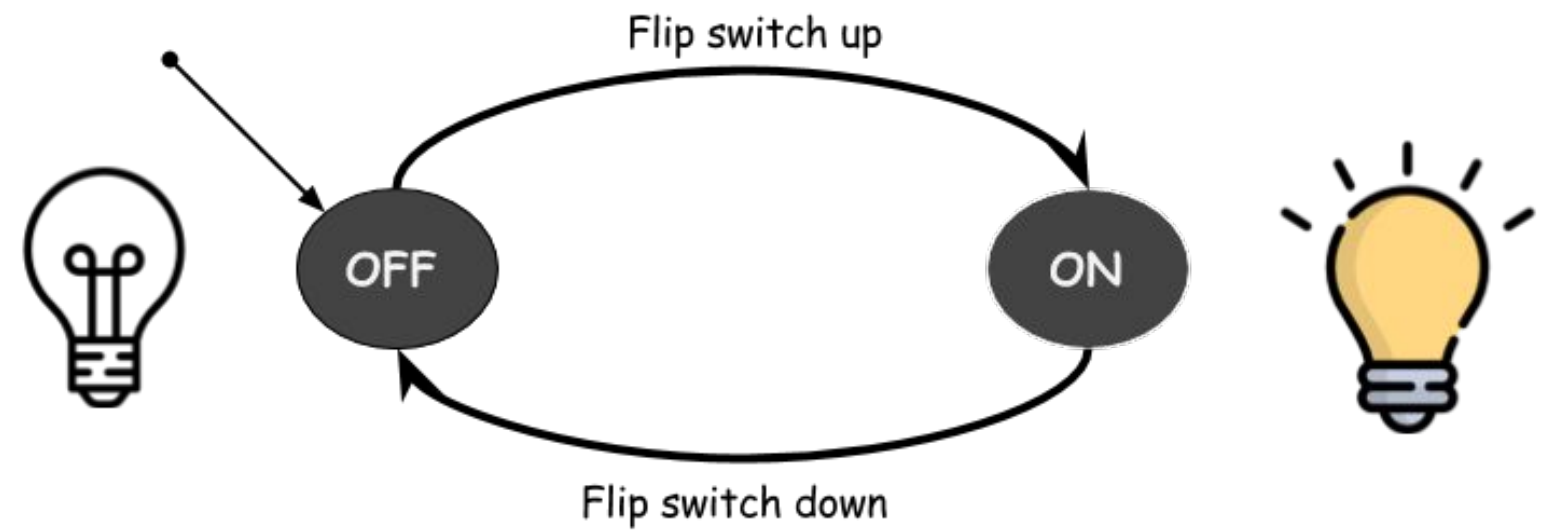
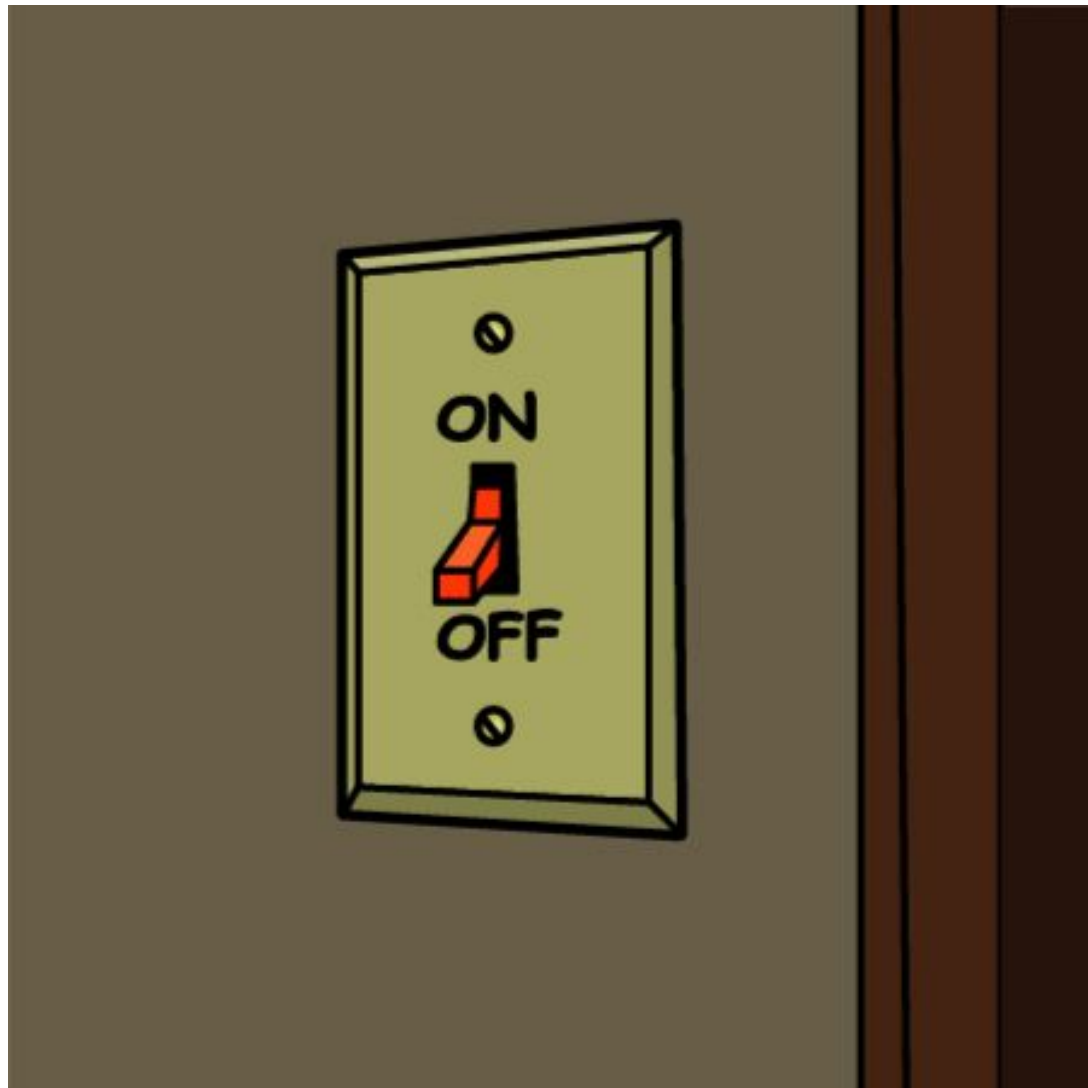
Finite-State Machines are used in :

- Text processing
- Compilers
- Network protocols
- Hardware design

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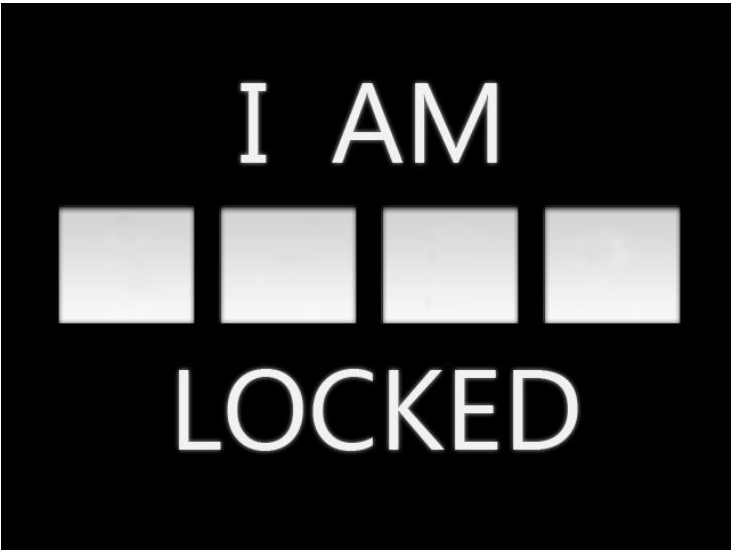
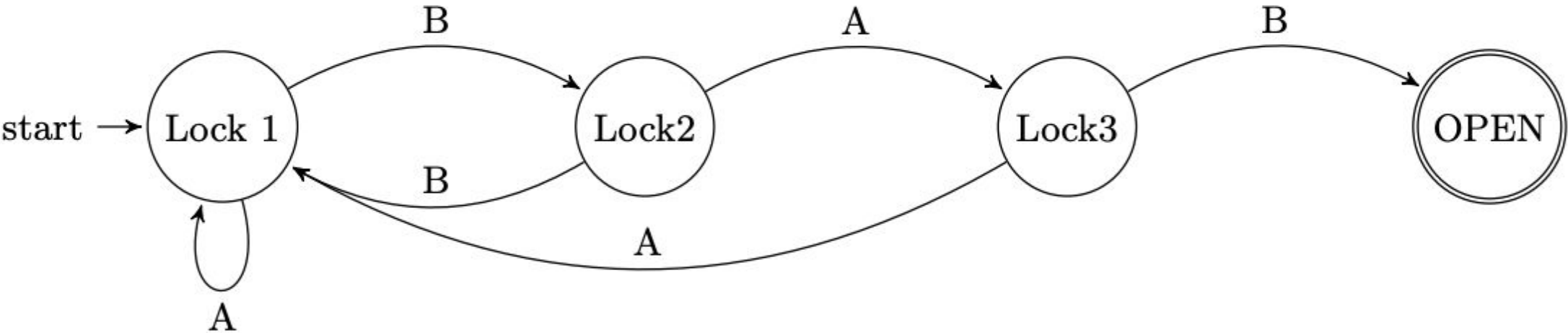
Switch Implementation



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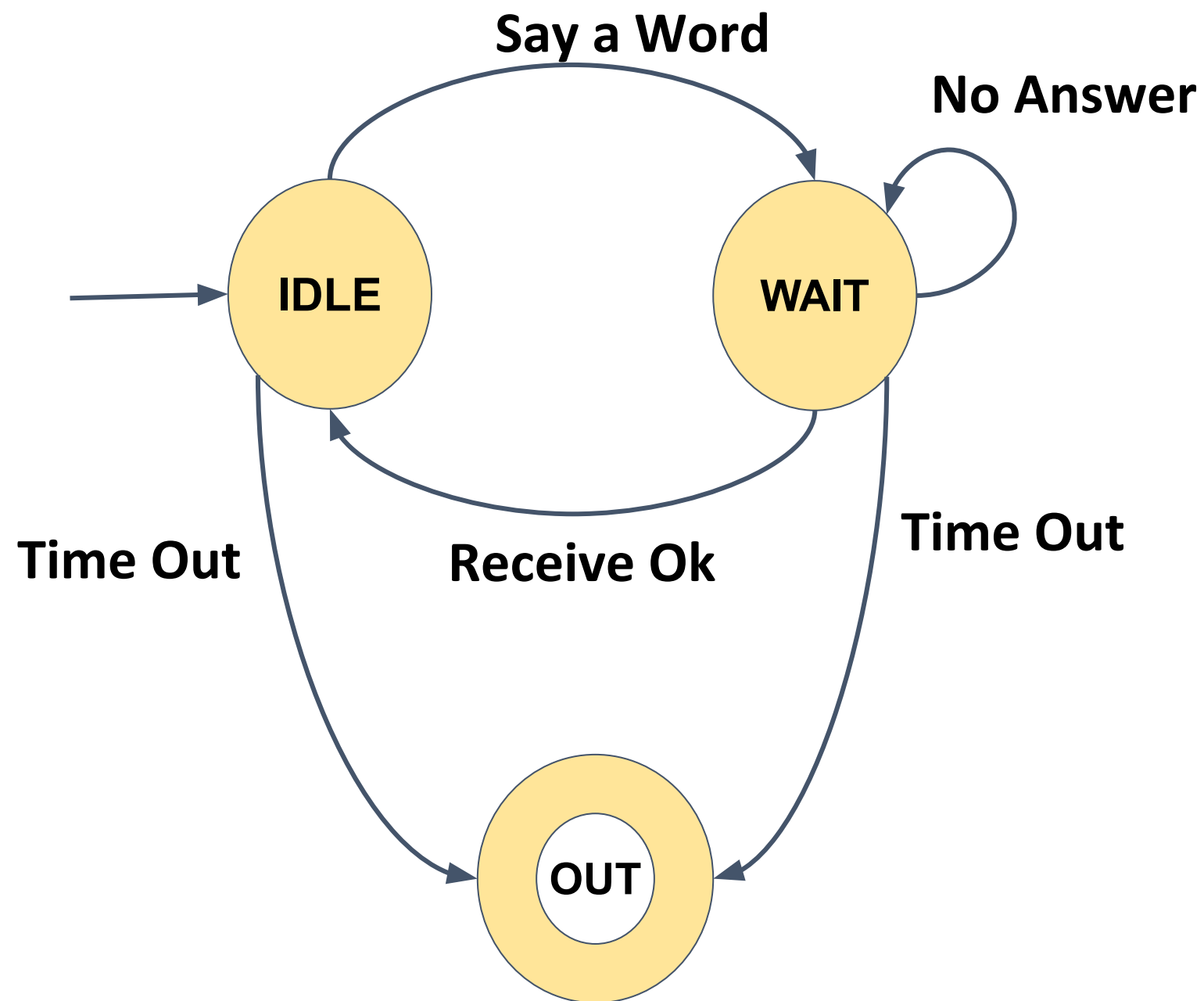
Code Lock Implementation



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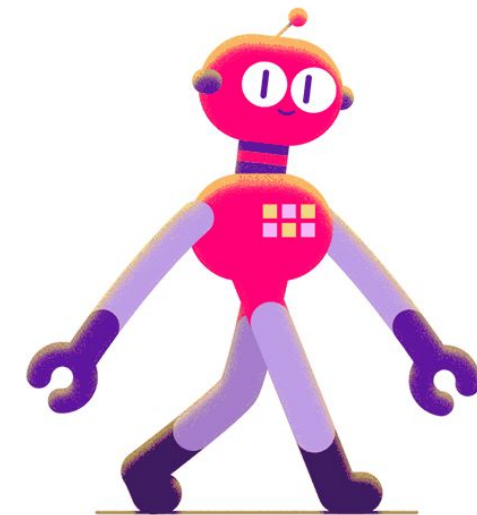
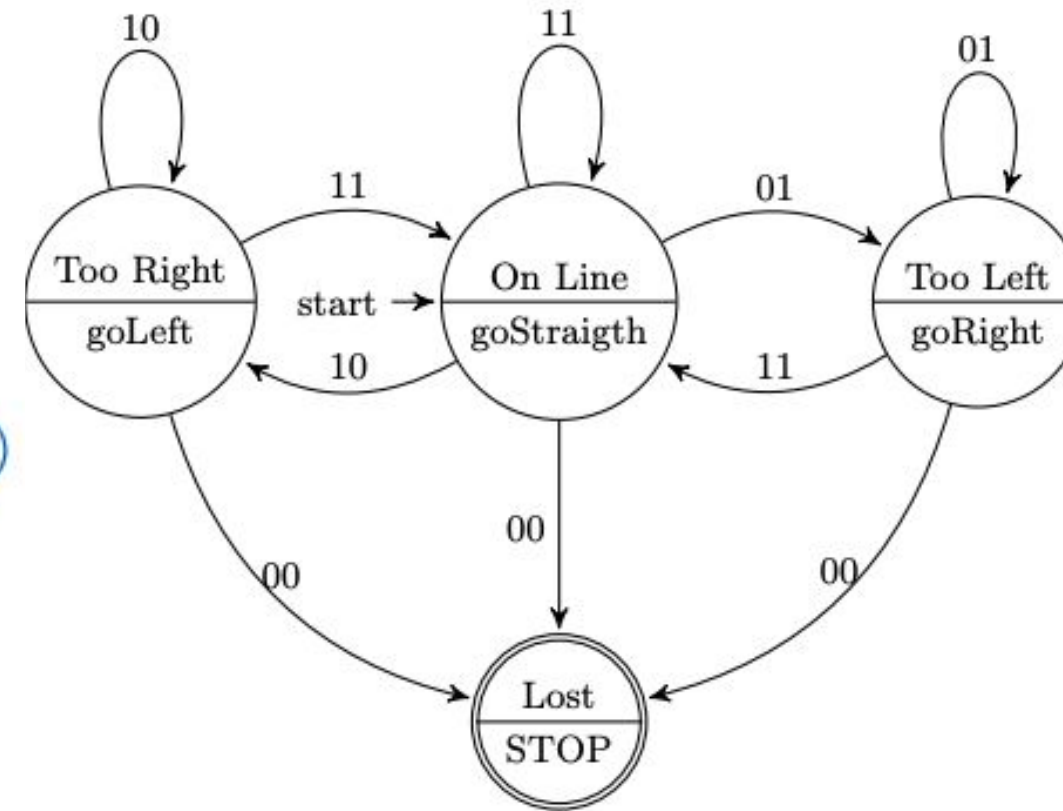
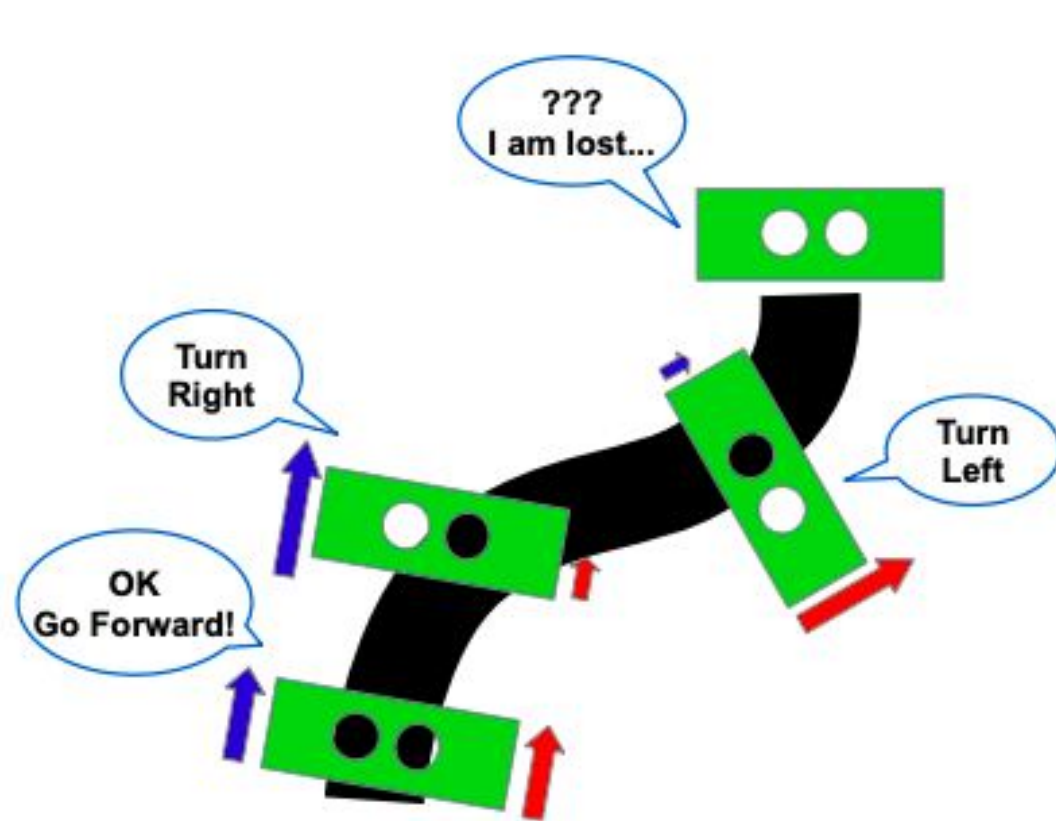
Communication Link



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Robotics



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Text Processing / Natural Language Processing

Spell Checker has two components :

1. **Error Detection (Detection of Misspelled word)**
2. **Spelling Correction or Suggestion Prediction**

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Error Detection in Spell Checker :

Create FSM for all the words in the dictionary

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Error Detection in Spell Checker :

Create FSM for all the words in the dictionary.

Let's say the dictionary is made up of the following words:

Henna

Elton

Glare

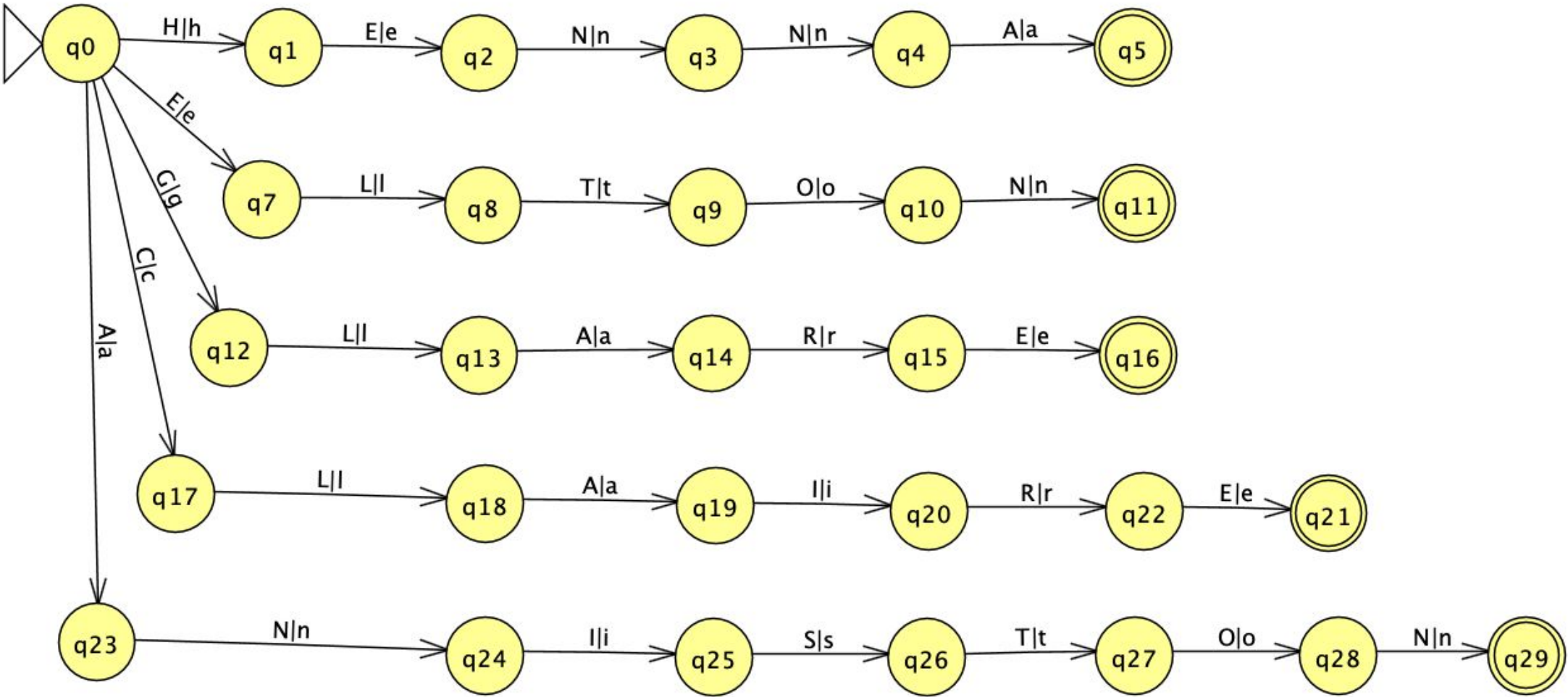
Claire

Aniston

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Error Detection in Spell Checker :
Our FSM will look like this:



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Spelling Correction in Spell Checker :

The word '**Helton**' is misspelled as it will not lead us to a final state. We must now provide a suggestion for correction.

We make use of **Edit Distance** in order to provide a possible correction

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Edit distance between two strings :

- Is the number of editing operations required to transform one into the other
 - Insertion
 - Deletion
 - Substitution
- Various Edit Distance Metrics are available, we will talk about **Levenshtein Distance**

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Levenshtein Distance

Levenshtein distance between two words is the minimum number of single-character edits (insertions, deletions or substitutions) required to change one word into the other.

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Levenshtein Distance = 0

T	E	S	T
T	E	S	T

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Levenshtein Distance = 2

T	E	S	T
T	E	A	M
		S	S

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Levenshtein Distance = 2

P	O	R	T	
P	A	R	T	Y

S

D

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Levenshtein Distance = 2

G	I		L	Y
G	E	E	L	Y

G		I	L	Y
G	E	E	L	Y

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Levenshtein Distance = 3

H		O	N	D	A	
H	Y	U	N	D	A	I

H	O		N	D	A	
H	Y	U	N	D	A	I

Spelling Correction in Spell Checker :

Misspelled Word is “Helton”

Let’s find Levenshtein Distance between “Helton” and other words in the dictionary:

Dictionary	L.V from “Helton”
Henna	4
Elton	1
Glare	5
Claire	6
Aniston	4

Spelling Correction in Spell Checker :

Misspelled Word is “Helton”

Hence our suggestion prediction is “Elton”

Dictionary	L.V from “Helton”
Henna	4
Elton	1
Glare	5
Claire	6
Aniston	4



THANK YOU

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